Azure DP300 Notes

Contents

Start Here

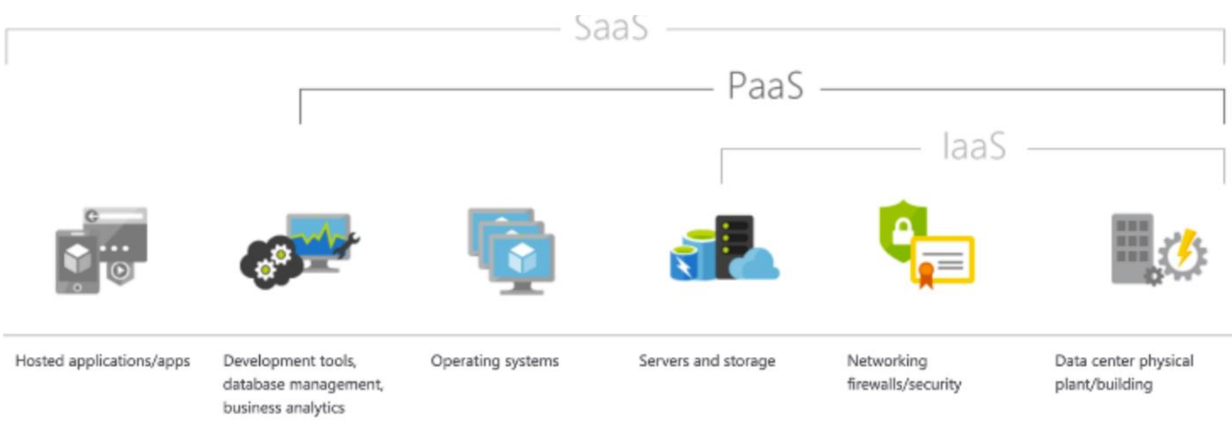
U R: Azure Cloud; Admin Consultant Sales Engineer SQL Data Design Optimize Support Person

Notes change often!

1. Exam DP-300: Administering Microsoft Azure SQL Solutions - Certifications | Microsoft Learn
2. Course DP-300T00--A: Administering Microsoft Azure SQL Solutions - Training | Microsoft Learn
3. DP-300 Exam Study Guide (microsoft.com)
4. MicrosoftLearning/dp-300-database-administrator: Repository for lab exercises and instructions for Microsoft DP-300 learning content (github.com)
5. DP-300T00-Administering-Relational-Databases-on-Azure (microsoftlearning.github.io)
6. Latest updates and version history for SQL Server - SQL Server | Microsoft Learn

Organization:

Off-Prem w/ Off-Prem Azure component; AD Users/Groups (resource access); SQL Server; SQL data (encrypting); storage; WLAN (networking/IP); VM; Service Level Agreement and Service Tier levels – (what comes in what package); Messaging; Apps & API's; IOT is everything else.



Some General Goals or Tasks:

1. Identifying which queries are consuming the most resources: Query Performance Insight - drill into an individual query, we will be able to see the query ID and the query itself, as well as the query aggregation type and associated time period. Furthermore, the query ID also correlates to the query ID located within the Query Store.

Tools:

Portal/SSMS/AZ Data Studio/VS Code/Execution Plans: Query Optimization (SARGArbility/Index/WHERE) Controlling the "search" behavior; Query Store.

Azure Resources/Availability Options (Zones/Regions/Failover per grouped product/Billing – DTU vs. vCore) RTO/RPO; Contained dbases & User Mapping; Scaling resources; SAS (Shared Access Signature)

Things to know about Resources

File locations SQL example:

General Purpose, tempdb is on direct-attached SSD, but the data and log files are in Azure Premium Storage.

How things work example:

- Cannot reserve the listeners IP address in Azure so you need to ensure something else does not come along and grab it otherwise there could be a conflict on the network, which in turn could cause availability headaches.
- Create/Manage Resources; Manage Access with AD; Disaster Recovery; Command line tools (AZ CLI, JSON, SQL, PS, Bash for starters)
- DMV to use and JOIN-ing them/Statistics-Metrics; Monitoring (Portal or otherwise)/Logs
- Relational databases perform best when executing set-based operations.
- Set-based operations perform data manipulation (INSERT, UPDATE, DELETE, and SELECT) in sets, where work is done on a set of values and produces either a single value or a result set.
- The alternative to set-based operations is to perform row-based work, using a cursor or a while loop. This type of processing is known as row-based processing, and its cost increases linearly with the number of rows impacted. That linear scale is problematic as data volumes grow for an application.
- One example of high availability would be an AG automatically failing over from one replica to another within an Azure region. That may take seconds, and at that point, you would need to ensure that the application can connect after the failover. SQL Server' s downtime would be minimal. A local RTO or RPO may potentially be measured in minutes depending on the critical nature of the solution or system.

Pattern-istic Thinking to avoid Complex-icating IT - Using Query Store

- The use of row-based operations with cursors or WHILE loops is important and there are other SQL Server anti-patterns that you should be able to recognize as well.
- Table-valued functions (TVF), particularly multi-statement table-valued functions, caused problematic execution plan patterns prior to SQL Server 2017.
- Using multi-statement table valued functions because
  - They can execute multiple queries within a single function
  - And aggregate the results into a single table.

A set of steps or "Workflow" in the correct order (example)

1. You discover that all Microsoft SQL Server Integration Services (SSIS) packages fail to run on the virtual machine.
2. Which four actions should you perform to resolve the issue?
   a. Actions: Attach the SSISDB database. 2. Turn on the TRUSTWORTHY property and the CLR property. 3. Open the master key for the SSISDB database. 4. Encrypt a copy of the master key by using the service master key.
3. The resources in App1Test must have the same configurations as the resources in App1Dev.

   a. Actions: 1. From the Azure portal, export the Azure Resource Manager templates. 2. Change the Server name and related variables in the templates. 3. From the Azure portal, deploy the templates. 4. From the database project, deploy the database schema and permissions.
4. You have an Azure Synapse Analytics dedicated SQL pool named Pool1
   a. and an Azure Data Lake Storage Gen2 account named Account1.
   b. Plan to access the files in Account1 by using an external table.
   c. Create a data source in Pool1 that you can reference when you create the external table.
   d. How should you complete the Transact-SQL statement?
5.

"Something" as a Service

PaaS; IaaS; SaaS; IoT (Platform, Infrastructure, Software – as a Service – Internet of Things (connect to devices – Hubs)

PaaS comes in several different service tiers. Each tier has varying capabilities, which provide a wide range of options when choosing this platform.

SQL "Family"

[High availability - Azure SQL Database and SQL Managed Instance | Microsoft Learn]

The **DTU** model is available in three different service tiers:

- Basic
- Standard
- Premium

You can purchase **vCore** databases in three different service tiers as well:

- General Purpose – This tier is for general purpose workloads. It is backed by Azure premium storage. It will have higher latency than Business Critical.
- Business Critical – This tier is for high performing workloads offering the lowest latency of either service tier. This tier is backed by local SSDs instead of Azure blob storage. It also offers the highest resilience to failure as well as providing a built-in read-only database replica that can be used to off-load reporting workloads.
- Hyperscale – Hyperscale databases can scale far beyond the 4 TB limit the other Azure SQL Database offerings and have a unique architecture that supports databases of up to 100 TB.

Auto-Failover Group is an availability feature that can be used with both Azure SQL Database and Azure SQL Database Managed Instance. Auto-Failover Groups let you manage how databases on an Azure SQL Database server or databases in Azure SQL Database Managed Instance are replicated to another region, and let you manage how failover could happen.

Virtual Machine Families

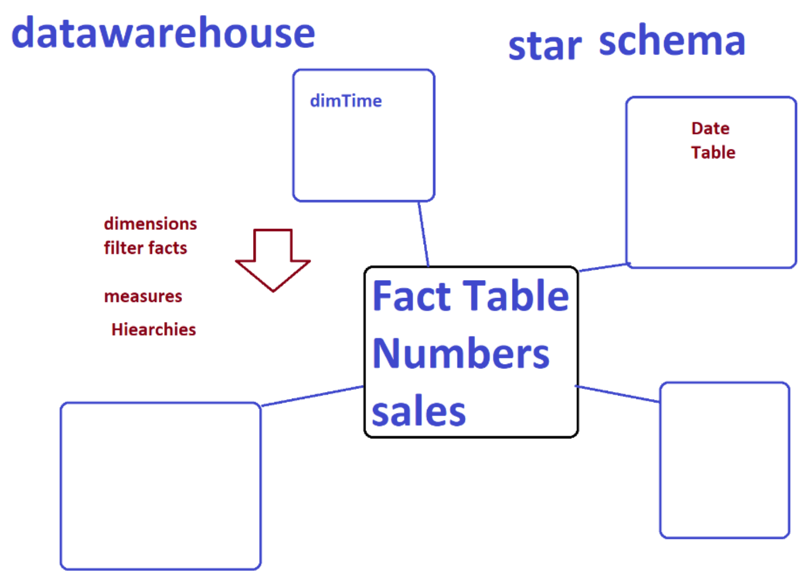
- General purpose – These VMs provide a balanced ration of CPU to memory. This VM class is ideal for testing and development, small to medium-sized database servers, and web servers with a low to medium amount of traffic.
- Compute optimized – Compute optimized VMs have a high CPU-to-memory ratio and are good for web servers with a medium amount of traffic, network appliances, batch processes, and application servers. These VMs can also support machine learning workloads that can't benefit from GPU-based VMs.
- Memory optimized – These VMs provide high memory-to-CPU ratio. These VMs cover a broad range of CPU and memory options (all the way up to 4 TB of RAM) and are well suited for most database workloads.
- Storage optimized – Storage optimized VMs provide fast, local, NVMe storage that is ephemeral. They are good candidates for scale-out data workloads such as Cassandra. It is possible to use them with SQL Server, however since the storage is ephemeral, you need to ensure you configure data protection using a feature like Always On Availability Groups or Log Shipping.
- GPU – Azure VMs with GPUs are targeted at two main types of workloads—naturally graphics processing operations like video rendering and processing, but also massively parallel machine learning workloads that can take advantage of GPUs.
- High performance compute – High Performance Compute workloads support applications that can scale horizontally to thousands of CPU cores. This support is provided by high-performance

CPU and remote direct memory access (RDMA) networking that provides low latency communications between VMs.

Components Of Automation

- Runbooks – Runbooks are the unit of execution in Azure automation.
    - Defined as one of three types: a graphical runbook based on PowerShell, a PowerShell script, or Python script. PowerShell runbooks are most commonly used to manage Azure SQL resources.
- Modules – Azure Automation defines an execution context for the PowerShell or Python code you are executing in your runbook. In order to execute your code, you need to import the supporting modules. For example, if you needed to run the Get-AzSqlDatabase PowerShell cmdlet, you would need to import the Az.SQL PowerShell module into your automation account.
- Credentials – Credentials store sensitive information that runbooks or configurations can use at runtime.
- Schedules – Schedules are linked to runbooks and trigger a runbook at a specific time.
- https://docs.microsoft.com/en-us/azure/automation/automation-intro

Star Schema



What is database sharding?

Sharding is a method for distributing a single dataset across multiple databases, which can then be stored on multiple machines.

This allows for larger datasets to be split into smaller chunks and stored in multiple data nodes, increasing the total storage capacity of the system.

Sharding is a partitioning pattern for the NoSQL age. It's a partitioning pattern that places each partition in potentially separate servers—potentially all over the world. This scale out works well for supporting people all over the world accessing different parts of the data set with performance.

How is sharding different from indexing?

Indexing is the process of storing the column values in a data structure like B-Tree or Hashing. It makes the search or join query faster than without index as looking for the values take less time. Sharding is to split a single table in multiple machines.

Why is sharding difficult in SQL?

Since data is distributed across many servers, a JOIN operation easily done in a single SQL database becomes difficult. Such an operation may hit many shards and require merging the responses. Shards may become unbalanced over time. In other words, some shards grow faster than others, thus becoming database hotspots.

There are four common sharding strategies:

- Horizontal or Range Based Sharding.
- Vertical Sharding.
- Key or hash based sharding.
- Directory based sharding.

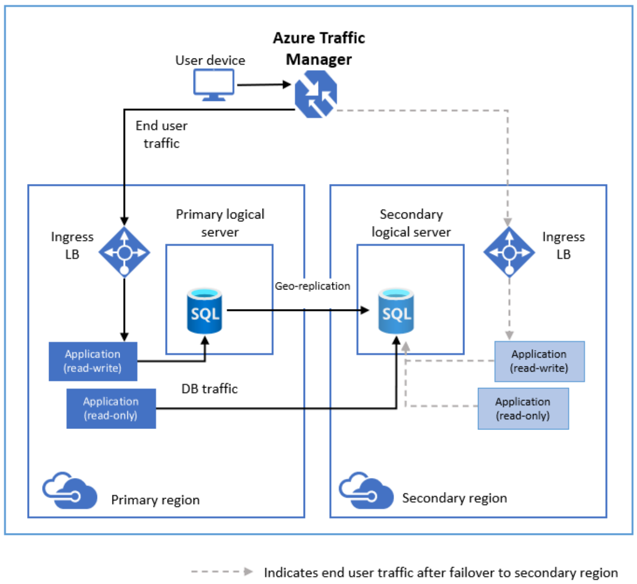
Key topics

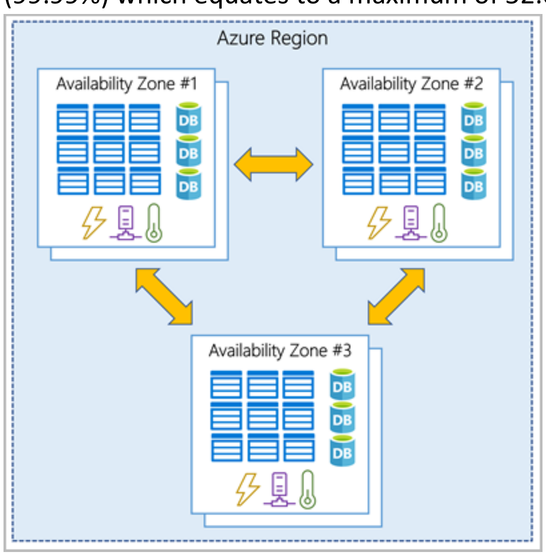
1. All service provisioning from ARM-JSON, Portal & GitHub Repo DevOps Pipeline with Bicep
2. View labs – time & allocation vs. trial length of 30 day s100.00
3. Data security in transit, rest, sensitivity
4. High Availability; Geo Redundancy; Zones & Scaling
5. Monitor and Optimize; Queries to IaaS
6. Migration; Automate; HADR
7. Data structures; indexing (DP900 is intro)
8. Key-value stores vs. graph databases

Big Bucket Items

9. Query Store etc… (areas and tools to manage them)

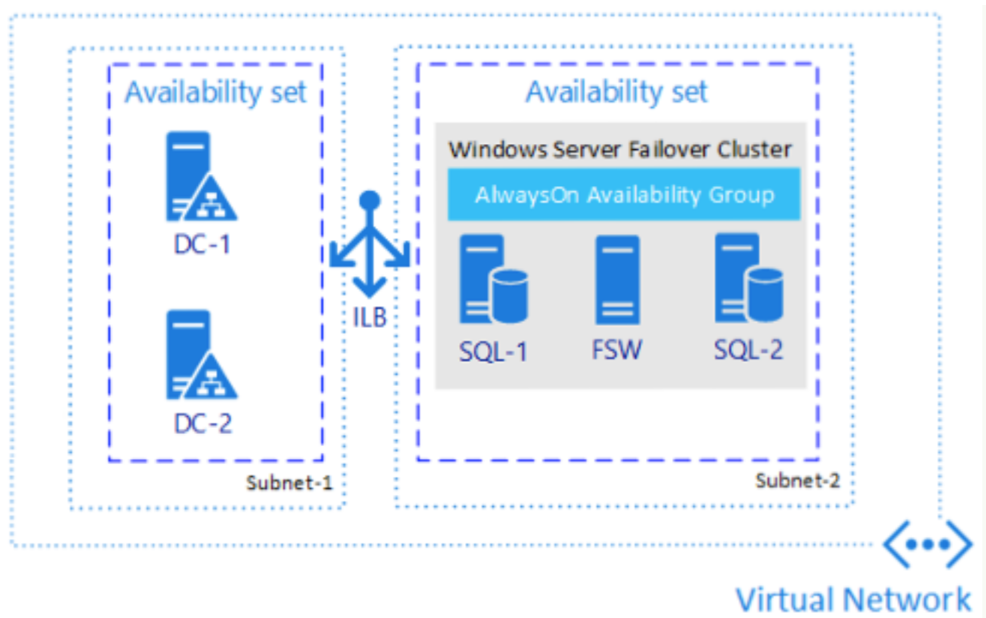
Grouping things

10. [Active geo-replication - Azure SQL Database | Microsoft Learn]() – good discussion
    a. Creates a replica of the database in another region that is asynchronously kept up to date.
    b. 

11.
12. Availability Zones
    a. In conjunction with your Azure virtual machines raises your uptime to four nines (99.99%) which equates to a maximum of 52.60 minutes of downtime per year.
    b. 

c.

13. Availability sets
    a. are similar to Availability Zones, except instead of spreading workloads across data centers in a region, they spread workloads across servers and racks in a data center.
    b. Since nearly all workloads in Azure are virtual, you can use availability sets in order to guarantee that the two VMs containing your Always On Availability Group members are not running on the same physical host.
    c. Availability sets can provide up to 99.95% availability, and should be used when Availability Zones are unavailable in a region, or an application cannot tolerate intra-zone latency.

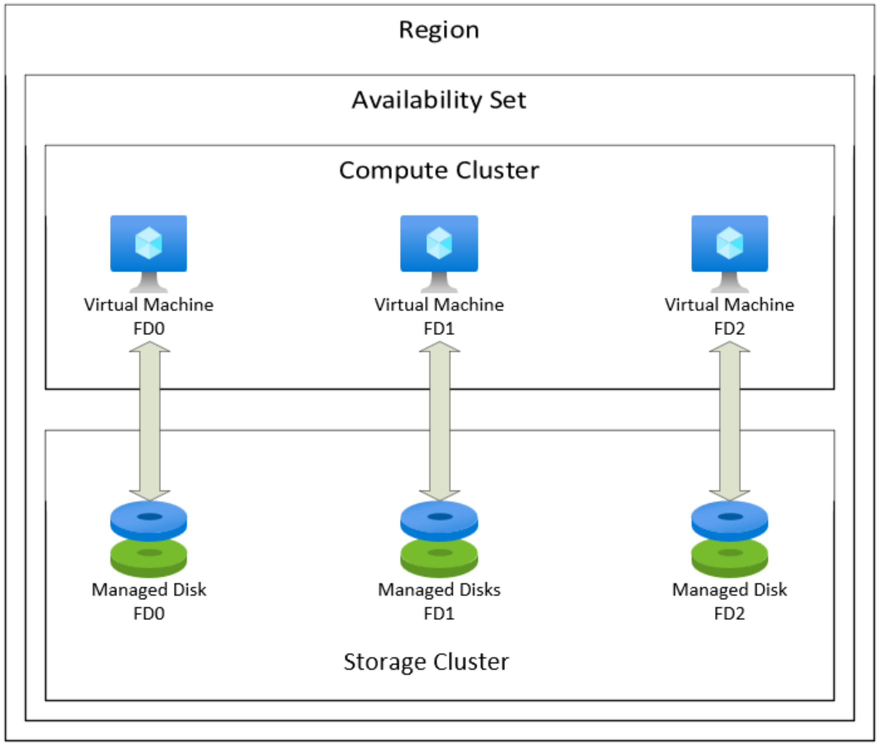


Availability sets are a logical grouping of VMs used to protect applications from hardware failures within an Azure data center

Availability zones, protect applications from complete Azure data center failures.

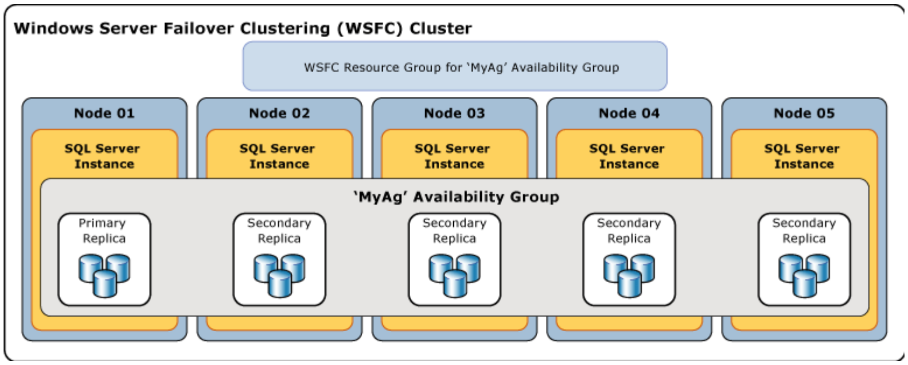
> An Availability group supports a replicated environment for a discrete set of user databases, known as availability databases.
>
> Deploying Always On availability groups for HA on Windows requires a Windows Server Failover Cluster (WSFC). Each availability replica of a given availability group must reside on a different node of the same WSFC.

d.



e.

14. Always On availability groups
    a. Can be implemented between two or more (up to a maximum of nine) SQL Server instances running on Azure virtual machines or across an on-premises data center and Azure.
    b. In an availability group, database transactions are committed to the primary replica, and then the transactions are sent either synchronously or asynchronously to all secondary replicas.
    c. The physical distance between the servers (that is, whether or not they are in the same Azure region) dictates which availability mode you should choose.
    d. Used for disaster recovery purposes.
        i. You can implement up to nine replicas of a database across Azure regions, and stretch this architecture even further using Distributed Availability Groups.
        ii. Availability Groups ensure that a viable copy of your database(s) is in another location beyond the primary region.
        iii. By doing so, you help to ensure that your data ecosystem is protected against natural disasters as well as some human made ones.
    e. The unit of failover is the group of databases and not the instance. While a failover cluster instance provides HA at an instance level, it doesn't provide disaster recovery.

f. 

15. SQL Server Failover Cluster Instance (FCI)
    a. To protect the entire instance, you could use a SQL Server Failover Cluster Instance (FCI), which provides high availability for an entire instance, in a single region.
    b. A FCI doesn't provide disaster recovery without being combined with another feature like availability groups or log shipping.
    c. FCIs also require shared storage that can be provided on Azure by using shared file storage or using Storage Spaces Direct on Windows Server.
    d. For Azure workloads, availability groups are the preferred solution for newer deployments, because the shared storage require of FCIs increases the complexity of deployments.
    e. However, for migrations from on-premises solutions, an FCI may be required for application support.
16. Disaster Recovery
    a. The two most important parameters of a good (or bad) BCDR plan are the recovery point objective (RPO) and recovery time objective (RTO)
    b. [RPO and RTO: What Are They and How to Calculate Them | Unitrends]

    Native SQL Server backups: SQL Server on an Azure virtual machine

        i. Use SQL agent jobs to back up directly to a URL linked to Azure blob storage.
        ii. Azure provides the option to use geo-redundant storage (GRS) or read-access geo-redundant storage (RA-GRS) to ensure that your backup files are stored safely across the geographic landscape.
    c. Azure Backup solution
        i. Requires an agent to be installed on the virtual machine.
        ii. The agent then communicates with an Azure service that manages automatic backups of your SQL Server databases.
    d. Azure Site Recovery
        i. A low-cost solution that will perform block level replication of your Azure virtual machine.
        ii. This service offers various options, including the ability to test and verify your disaster recovery strategy.
        iii. This solution is best used for stateless environments (for example, web servers) versus transactional database virtual machines.
        iv. Supported for use with SQL Server, but keep in mind that you will need to set a higher recovery point which means potential loss.

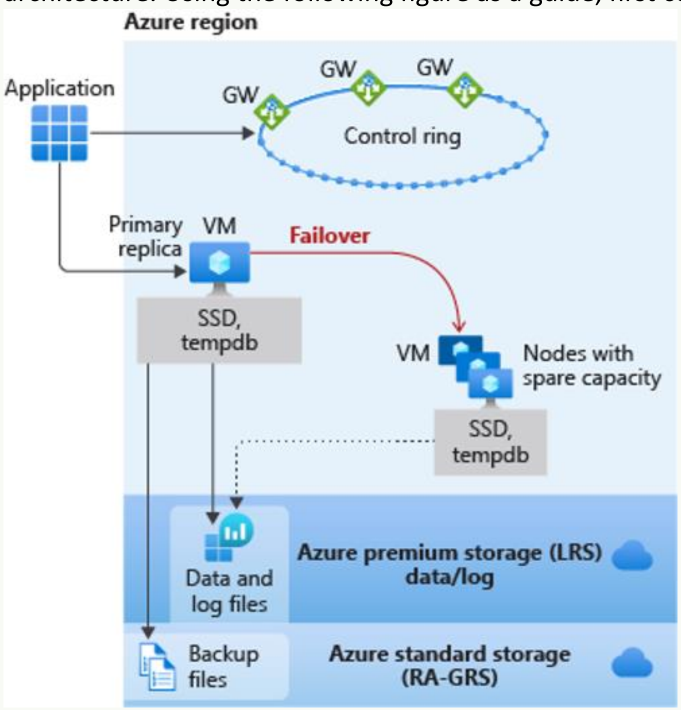
v. In this case, your RTO will essentially be your RPO.

17. Memory per core: The amount of memory (RAM) per CPU core. If a compute node has 2 CPUs, each having 6 cores and 24GB (gigabytes) of installed RAM, then this compute node will have 2GB of memory per core. Memory per node: The total amount of installed RAM in a compute node.
18. Failover groups
    a. Are built on top of the technology used in geo-replication, but provide a single endpoint for connection.
    b. The major reason for using failover groups is that the technology provides endpoints, which can be utilized to route traffic to the appropriate replica.
    c. Your application can then **connect after a failover without connection string changes**.
19. Azure SQL Database Hyperscale
    a. Changes the paradigm and allows for databases to be 100 TB or more.
    b. Introduces new horizontal scaling techniques to add compute nodes as the data sizes grow.
    c. The cost of Hyperscale is the same as the cost of Azure SQL Database; however, there's a per terabyte cost for storage.
    d. Note that once an Azure SQL Database is converted to Hyperscale, you can't convert it back to a "regular" Azure SQL Database.
    e. Hyperscale is the ability for an architecture to scale appropriately as demanded.
    f. Scale up the primary compute size in terms of resources like CPU and memory, and then scale down, in constant time.
        i. Storage is shared, scaling up and scaling down isn't linked to the volume of data in the database.
    g. In/Out: use extra compute replicas as read-only replicas to offload your read workload from the primary compute. In addition to read-only, these replicas also serve as hot-standbys to failover from the primary.

Blueprints – Recognize the Service Tier from the "Parts"

General Purpose

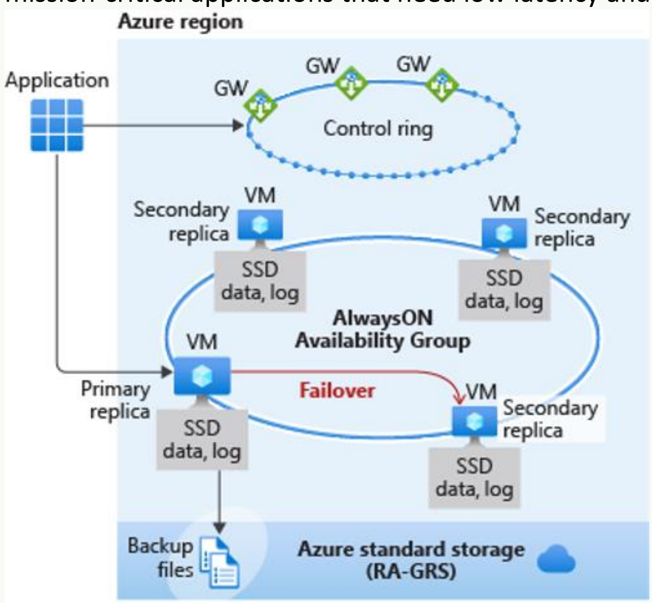
1. Databases and managed instances in the General Purpose service tier have the same availability architecture. Using the following figure as a guide, first consider the application and control ring.



2.
3. The application connects to the server name, which then connects to a gateway (GW) that points the application to the server to connect to, running on a VM.
    a. With General Purpose, the primary replica uses locally attached SSD for the tempdb.
    b. The data and log files are stored in Azure Premium Storage, which is locally redundant storage (multiple copies in one region).
    c. The backup files are then stored in Azure Standard Storage, which is RA-GRS by default. - globally redundant storage (with copies in multiple regions).
4. All of Azure SQL is built on Azure Service Fabric, which serves as the Azure backbone.
    a. If Azure Service Fabric determines that a failover needs to occur, the failover will be similar to that of a failover cluster instance (FCI).
    b. The service fabric will look for a node with spare capacity and spin up a new SQL Server instance.
    c. The database files will then be attached, recovery will be run, and gateways will be updated to point applications to the new node.
    d. No virtual network or listener or updates are required. This capability is built in.

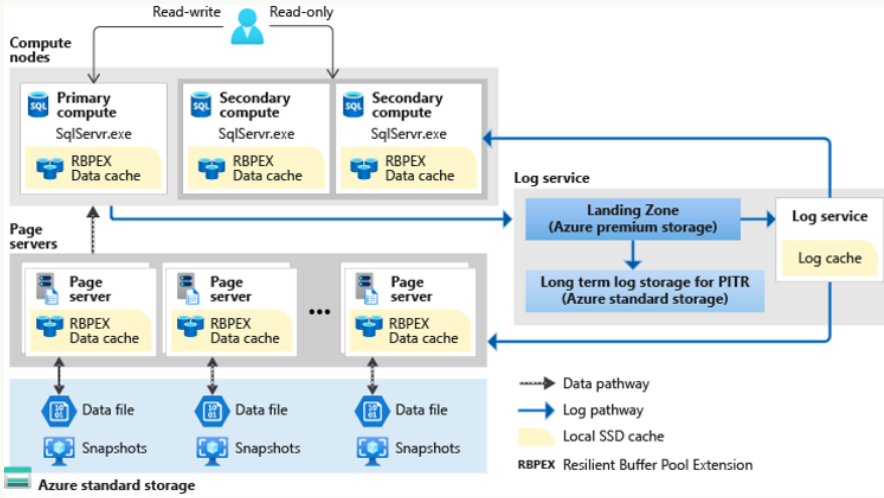
Business Critical

5. Business Critical can generally achieve the highest performance and availability of all Azure SQL service tiers (General Purpose, Hyperscale, Business Critical). Business Critical is meant for mission-critical applications that need low latency and minimal downtime.



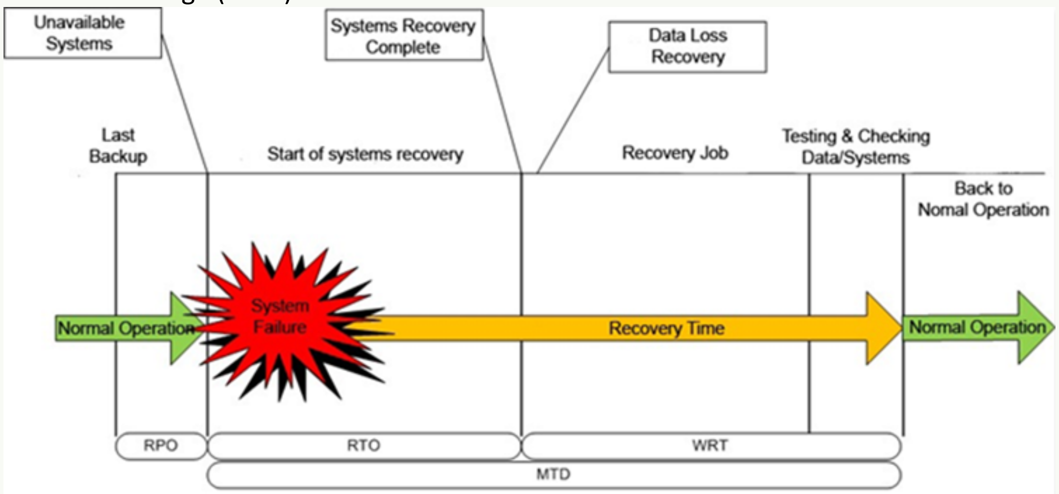
6.
7. Using Business Critical is like deploying an Always On Availability Group (AG) behind the scenes.
8. Unlike in the General Purpose tier, in Business Critical,
    a. Data and log files are all running on direct-attached SSD, which significantly reduces network latency. (General Purpose uses remote storage.)
    b. In this AG, there are three secondary replicas.
    c. One of them can be used as a read-only endpoint (at no additional charge).
    d. A transaction can complete a commit when at least one of the secondary replicas has hardened the change for its transaction log.
9. Read scale-out with one of the secondary replicas supports session-level consistency.
    a. So if the read-only session reconnects after a connection error caused by replica unavailability, it might be redirected to a replica that's not 100% up to date with the read-write replica.
    b. Likewise, if an application writes data by using a read-write session and immediately reads it by using a read-only session, the latest updates might not immediately be visible on the replica.
    c. The latency is caused by an asynchronous transaction log redo operation.
10. If any type of failure occurs and the service fabric determines a failover needs to occur,
    a. Failing over to a secondary replica is fast because the replica already exists and has the data attached to it. In a failover, you don't need a listener.
    b. The gateway will redirect your connection to the primary even after a failover.
    c. This switch happens quickly, and then the service fabric takes care of spinning up another secondary replica.

Hyperscale

11. 

12. The Hyperscale service tier is available only in Azure SQL Database. This service tier has a unique architecture.
13. Because this architecture uses paired page servers, you can scale horizontally to put all the data in caching layers.
    a. This new architecture also allows Hyperscale to support databases as large as 100 TB.
    b. Because it uses snapshots, nearly instantaneous database backups can occur regardless of size.
    c. Database restores take minutes rather than hours or days.
    d. You can also scale up or down in constant time to accommodate your workloads.
14. It's interesting to note how the log service was pulled out in this architecture.
    a. The log service is used to feed the replicas and the page servers.
    b. Transactions can commit when the log service hardens to the landing zone.
    c. So the consumption of the changes by a secondary compute replica isn't required for a commit.
    d. Unlike in other service tiers, you can determine whether you want secondary replicas.
    e. You can configure zero to four secondary replicas, which can all be used for read-scale.
15. As in the other service tiers, an automatic failover will happen if service fabric determines it needs to.
    a. But the recovery time will depend on the existence of secondary replicas.
    b. For example, if you don't have replicas and a failover occurs, the scenario will be similar to that of the General Purpose service tier: the service fabric first needs to find spare capacity.
    c. If you have one or more replicas, recovery is faster and more closely aligns to that of the Business Critical service tier.
16. Business Critical maintains the highest performance and availability for workloads with small log writes that need low latency.
    a. But the Hyperscale service tier allows you to get a higher log throughput in terms of MB/second. So you'll need to consider your workload when you choose between the two.

17. The Maximum Tolerable Downtime (MTD) is the amount of time that a business may be without a service before irreparable harm occurs. The measure is sometimes also called the Maximum Tolerable Outage (MTO).



18.

Vocab and Other What Not

1. Availability Sets; Groups; Zones etc… (see Blueprints)
2. Deltastore
3. Application that requires access to the operating system (OS)
4. vCore database service tiers
5. Purchasing models

Tools you can use to map your server estate and identify compatibility with your target Azure platform

6. Azure Migrate Tool
7. MAP Toolkit
    a. The Microsoft Assessment and Planning Toolkit automatically gathers an inventory of all of the SQL Servers in your network, gathering version and server information and providing reports on the information gathered.
8. Database Experimentation Assistant:
    a. This tool can be used to evaluate version upgrades of SQL Server by checking syntax compatibility and provides a platform to evaluate query performance on the target version.

More vocabulary

9. SQL Server Resource Provider for Azure VMs
10. self-service restore
11. Purchase model: DTU vs. vCore
12. Query Store catalogue view
13. OAuth 2.0 is the industry protocol for authorization.
14. Microsoft SQL Server Data Tools (SSDT)
15. Hybrid transactional analytic processing (HTAP), where data is being loaded into the underlying table, and at the same time reports are being run on the table. By filtering the index (typically on a date field), this design allows for both good insert and reporting performance.
16. Azure SQL Database - is the only deployment option with a possible SLA of 99.995 percent. The customer can obtain that SLA by using the Business Critical service tier and enabling availability zones.
17. Azure Storage - is a highly scalable, secure storage platform that offers a range of solutions to meet the needs of many applications. The following types of storage support encryption at rest with either a Microsoft managed or a user-defined encryption key.
18. Blob Storage – Blob storage is what is known as object-based storage and includes cold, hot, and archive storage tiers. In a SQL Server environment, blob storage will typically be used for database backups, using SQL Server' s back up to URL functionality.
19. File Storage – File storage is effectively a file share that can be mounted inside a virtual machine, without the need to set up any hardware. SQL Server can use File storage as a storage target for a failover cluster instance.
20. Disk Storage – Azure managed disks offer block storage that is presented to a virtual machine. These disks are managed just like a physical disk in an on-premises server, except that they are virtualized. There are several performance tiers within managed disks depending on your

workload. This type of storage is the most commonly used type for SQL Server data and transaction log files.

21. Azure Resource Manager templates are a declarative deployment approach that describes the desired structure and state of the resources to be deployed
22. Declarative programming is a paradigm describing WHAT the program does, without explicitly specifying its control flow. Imperative programming is a paradigm describing HOW the program should do something by explicitly specifying each instruction (or statement) step by step, which mutate the program's state.
23. Workload Group
24. Accelerated Database Recovery (ADR)
25. COPY_ONLY backup since Azure SQL Database Managed Instance is maintaining the log chain
26. Declarative model, which defines what should be created.
    a. Azure Resource Manager templates
27. Imperative frameworks
    a. PowerShell - Azure CLI – a prescriptive order of tasks to be executed.
28. Infrastructure as code: all resources are defined a set of scripts that are stored in source control and can easily be deployed to a new environment. (DevOps, Pipelines, Github)
29. Always Encrypted keys with role separation.
    a. Only stores encrypted data in the SQL Server so the data is never visible to a user of the database without the column key, held by the application.
    b. The security admin doesn't access the database, and the DBA doesn't access the physical keys in plaintext. You can also use SQL Audit, Data Classification, and Dynamic Data Masking in combination to monitor.
30. Execution Plan
31. Lightweight Profiling: use the USE HINT query hint with QUERY_PLAN_PROFILE to enable lightweight profiling at the query level and get Actual Execution Plan
32. Azure Backup
33. Storage Spaces Direct
34. Azure Shared Disk
35. Point-in-Time Restore (PITR)
36. Database normalization
37. Data Lake Store
38. Log Analytics
39. Monitor
40. Event Hub
41. HDInsight
42. Azure Storage
43. GracePeriodWithDataLossHours
44. Actual versus estimated execution plans can be confusing. The difference is that the actual plan includes runtime statistics that are not captured in the estimated plan.
45. The basic storage unit in SQL Server is the Page, with each page size equal to 8KB. And each 8 pages with 64KB size called Extent.
46. Avro format
47. Kusto Query Language

48. IOPS means Input/Output Per Second, and it' s absolutely not referring to bandwidth. There is a tight relation between bandwidth and IOPS, but we need another parameter to do that. The Input/output size.
49. SOS_SCHEDULER_YIELD
50. Azure PostgreSQL also offers a scale-out hyperscale solution called Citus.
    a. Citus provides both scale-out and additional high availability for a server group. If enabled, a standby replica is configured for every node of a server group, which would also increase cost since it would double the number of servers in the group.
51. Randsomeware attack
52. End_list

Code Editing Method + reason for use:

Runbook vs. Ledger vs. Python Notebook

Concept	Editing Tool / Method	General Reason for Use	Typical User
Runbook	- Edited in **Azure Automation Portal**, PowerShell/CLI, or YAML/JSON files in a DevOps repo. - Can be authored in code editors (VS Code) if using scripts.	- Automates **operational tasks** (backups, failovers, scaling). - Provides repeatable instructions so admins don't rely on memory. - Can run unattended in scheduled workflows.	SysAdmins, Cloud Engineers, DevOps
Ledger	- Edited in **databases** (append-only storage), blockchain tools, or structured record systems. - Generally *not "edited"* but **appended** (immutable design).	- Tracks **transactions or events** with integrity. - Ensures auditability and prevents tampering. - Used in finance, compliance, or blockchain scenarios.	Accountants, Compliance Officers, Blockchain Devs
Python Notebook	- Edited in **Jupyter Notebook**, VS Code, Databricks, or Azure ML Studio. - Mix of **Python code cells + Markdown**.	- Interactive coding for **data analysis, machine learning, and visualization**. - Supports experimenting, documenting, and sharing results in one place.	Data Scientists, Analysts, Students

🔑 **Key Distinctions**

- **Runbook = Automation** → like a recipe for IT operations.
- **Ledger = Record-Keeping** → like an official logbook, designed to be tamper-proof.
- **Python Notebook = Exploration & Analysis** → like a lab notebook for coding and data experiments.

SQL Server Disaster Recovery Backup Plan

Aligns with DP-300 exam expectations, with a **5-minute Point-in-Time (PIT) Recovery Goal**. Let's break it into **frequency, method, and management approach**.l

---

🔷 Common Backup Frequency for PIT ≤ 5 Minutes

In SQL Server (on-prem or Azure VM/MI):

Backup Type	Common Frequency	Why
Full Backup	1x per day (often off-peak hours)	Establish full baseline
Differential Backup	Every 4–6 hours	Reduces restore chain length (smaller than full, faster than logs)
Transaction Log Backup	Every 5 minutes (or even 1 min for critical systems)	Supports **point-in-time restore** down to last 5 min (meets PIT objective)

👉 With this schedule:

- Restore = Last full → Latest differential → All subsequent log backups until failure.
- PIT granularity = 5 minutes (depending on log backup frequency).

🔷 Preferred Method (DP-300 Mindset)

1. SQL Server on Azure VM / Managed Instance

- Use **SQL Agent Jobs** or **Automated Runbooks (Azure Automation)**.
- T-SQL or maintenance plan to schedule backups:
- BACKUP DATABASE MyDB TO URL='...' WITH INIT, STATS=10;
- BACKUP LOG MyDB TO URL='...';
- Store backups in **Azure Blob Storage** for durability.
- Consider **Azure Backup integration** for VM-level protection.

---

2. Azure SQL Database (PaaS)

- No manual backup jobs needed.
- Azure automatically:
    - Full backup: weekly

  - Differential: every 12 hours
  - Log backups: every 5–10 minutes
- These meet the PIT goal **out-of-the-box** (no setup).
- Restore via **Portal or PowerShell** → select PIT.

🔷 Task, Plan, Runbooks (Training Level)

Step	Task	Tool/Method
Plan	Define RPO (Recovery Point Objective) = 5 min, RTO (Recovery Time Objective).	Document in DR Runbook.
Full Backup	Schedule daily full backup to Azure Blob Storage.	SQL Agent Job or Azure Automation Runbook.
Differential Backup	Every 4–6 hours.	Same job/runbook, separate schedule.
Log Backup	Every 5 minutes.	SQL Agent Job or Runbook, with retry logic.
Test Restore	Quarterly drill: restore to a dev/test instance.	T-SQL or SSMS.
Automation	Store jobs in **Azure Automation Runbook** for centralized management.	Runbook integrates with Key Vault (for credentials).

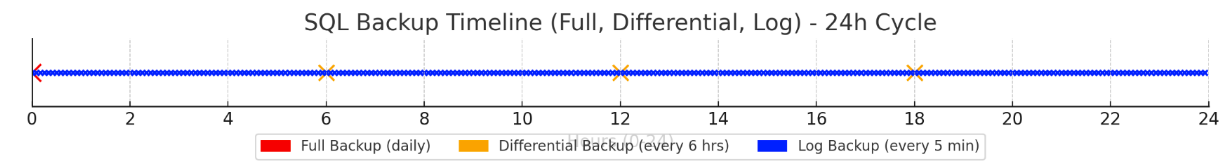
🔷 Cheat-Sheet View (DP-300 Exam Prep)

- **IaaS / MI:** You design/schedule backups (Full/Diff/Log).
- **PaaS:** Azure handles backups; you configure retention + restore.
- **PIT Objective:** Controlled by **log backup frequency** (set to 5 min).
- **Preferred Method:**
  - For training → **Runbooks** (DP-300 loves "automation + repeatability").
  - For VM → **SQL Agent Jobs** are acceptable.
  - For PaaS → Use built-in **Auto-backup service**.

✅ **Training-Friendly Plan**:

- Full = Daily @ midnight
- Diff = Every 6 hrs
- Logs = Every 5 min
- Automated via **Azure Automation Runbooks (preferred)** or SQL Agent.
- Test restores regularly.

Timeline diagram (24-hr backup cycle showing full/diff/log cadence)



Here's the **24-hour backup timeline**:

- 🔴 **Full backup** once a day (midnight).
- 🟠 **Differential backups** every 6 hours (06:00, 12:00, 18:00).
- 🔵 **Transaction log backups** every 5 minutes (dense dots across the day).

This cadence gives you a **5-minute PIT restore capability** while keeping restore chains manageable.

What's merged, What's Required at Restore, and the Tail-Log Question

Excellent drill-down 👌 — this is exactly the type of reasoning DP-300 exam designers like to test. Let's break down **what's merged, what's required at restore, and the Tail-Log question**.

🔷 Backup Data Content (Merge Between Steps)

Backup Type	Contains	Relation to Previous Backups
Full	Entire database at that moment (all data + part of log for consistency)	Fresh baseline — stands alone
Differential (Diff)	All extents **changed since the last Full**	Cumulative: each new Diff is larger as the day progresses (not incremental chain)
Transaction Log	All committed + uncommitted changes since the last Log backup	Chain: each log backup depends on all prior log backups since last Full (or Diff)

👉 So:

- **Full → everything** (biggest file).
- **Diff → "delta since last Full"** (not delta since last Diff).
- **Log → "since last log"**, smallest units but requires chaining.

🔷 Restore Sequence Requirements

Let's assume:

- Full at midnight
- Diff at noon
- Logs every 5 min

Scenario: Crash at 14:05

1. Restore **last Full** (midnight).
2. Restore **last Diff** (noon).
3. Restore **all Log backups** since that Diff (12:05, 12:10 … up to 14:05).

➡️ You do **not just need the last log** — you need the chain of logs since the Diff.
Each log backup "fills in" the timeline up to the next one.

🔷 **Tail-Log Backup**

- A **Tail-Log backup** captures log records **not yet backed up** at the moment of failure (the "tail end").
- This is required for **zero-data-loss restores** if the log file is still accessible.
- Syntax:
- BACKUP LOG MyDB TO DISK = 'tail.trn' WITH NO_TRUNCATE;
- Without it, you can only restore up to the last successful log backup.

🔷 **Can Tail-Log Backup Be Automated?**

- **Yes, partially:**
  - Normally Tail-Log is a *manual DBA action* during recovery because you run it right after failure detection.
  - However, you can **script it in a Runbook / SQL Agent Job** as part of a "Failure Detection → Emergency Script" workflow.
  - In practice, true *automation* is tricky because Tail-Log assumes DB is damaged/offline and you intervene before restore.

➡️ In cloud-friendly strategies, many teams rely on **5-minute log backup + HA/replication/geo-replica** instead of Tail-Log automation.

🔷 **DP-300 Exam-Friendly "Cheat Answer"**

- **Full backup:** complete dataset.
- **Diff backup:** cumulative changes since last Full.
- **Log backup:** changes since last Log; restore requires the **entire log chain**, not just the last one.
- **Tail-Log:** needed at restore time for zero data loss; usually manual, though can be scripted into Runbooks.

Restore Chain Diagram:

- 🟥 **Full backup** (midnight baseline).
- 🟧 **Differential backup** (all changes since last Full).
- 🔵 **Log backups** (sequential chain — every 5 min).
- ⚡ **Crash** (at 14:05).
- 🟩 **Tail-log backup (optional)** — captures last unbacked changes before restore.

➡️ During restore: **Full → Diff → all Logs (since Diff) → Tail-log if available**.

Where Are the Backup Files

🔷 **Backup File (.bak) in Azure Context**

1. SQL Server on Azure VM (IaaS)

- Works **exactly like on-prem SQL Server**.
- .bak files are created and stored wherever you choose:
  - Local disk inside the VM.
  - Azure Blob Storage via **BACKUP TO URL** (container in a storage account).
- **You control** the backup/restore using T-SQL or SSMS.
- You can **restore from that .bak** directly with RESTORE DATABASE FROM URL.

👉 In this model, yes, you can see and pick your .bak file from storage (if using SSMS or scripting). But the Azure Portal itself won't show a ".bak list" — you browse storage blobs.

---

2. Azure SQL Managed Instance

- Managed Instance supports .bak files, but **only from/to Azure Blob Storage** (via **URL**).
- You must upload the .bak file to a container and grant an **SAS token**.
- Then you restore with T-SQL (RESTORE DATABASE FROM URL).
- Again: Portal doesn't list .bak files; you specify path + SAS.

---

3. Azure SQL Database (PaaS, single database or elastic pool)

- ⚠️ **No direct .bak support.**
- Instead, backups are handled automatically by Azure:
  - Full backups weekly, differential every 12 hours, log backups every 5–10 minutes.
  - Stored in **RA-GRS Azure storage (geo-redundant)** — but you don't see or manage the .bak.
- Restore is done in the Portal by choosing:
  - **Point-in-time restore** (drop-down list with timestamps).
  - **Geo-restore** (from backup in paired region).

👉 Here, the portal **does** show restore options in a dropdown — but it's **timestamps of restore points**, not .bak files.

🔷 **Summary Table**

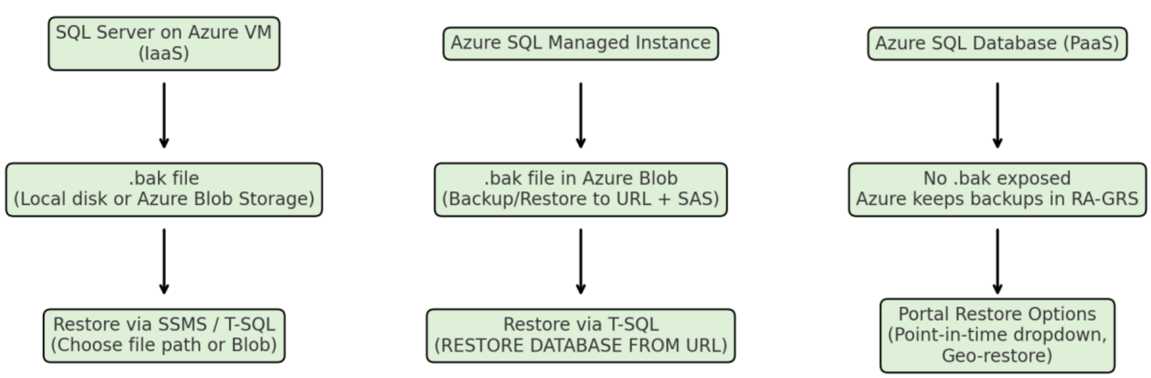
Deployment	Backup File (.bak)?	Where Stored	Restore Experience
SQL on VM (IaaS)	Yes	VM disk or Blob via URL	SSMS/Portal (from file path or blob)
Managed Instance	Yes (via URL only)	Azure Blob Storage	T-SQL with URL + SAS token
Azure SQL Database (PaaS)	No	Azure-managed (RA-GRS storage)	Portal dropdown (timestamps, not .bak files)

✅ **So the short answer:**

- If you're using **IaaS SQL or Managed Instance**, .bak files live in **Azure Blob Storage** when doing **BACKUP TO URL**, and you restore by pointing to that file (no portal dropdown list of files).
- If you're using **Azure SQL Database (PaaS)**, you never touch a .bak file — instead you get **restore points in a dropdown** inside the portal.

Diagram 3 Azure SQL Backup/Restore Flows

- **VM (IaaS)**: .bak on disk or Blob → restore via SSMS/T-SQL.
- **Managed Instance**: .bak in Blob (URL + SAS) → restore via T-SQL.
- **PaaS (Azure SQL DB)**: No .bak → restore from portal restore points dropdown.



SQL Migration Workflows

Across **data vs. files** with the tools, the main workflow steps, and (where relevant) the **billing/cost model**.

SQL Migration Workflows (DP-300 Point of View)

Scenario	Tool / Service	Workflow Summary	Best For	Cost / Billing Unit
Lift & Shift DB	**Azure Migrate** + **Azure Database Migration Service (DMS)**	Assess existing SQL Server → Provision Azure SQL (MI/DB) → Use DMS to migrate schema + data	Full SQL Server migration with minimal downtime	**Free** for SQL migration, but Azure SQL target incurs **vCore or DTU billing**
Schema-First Migration	**SQL Server Data Tools (SSDT)** or **Azure Data Studio Migration Extension**	Extract schema → Deploy to target → Load data separately	When schema redesign or versioning is needed	Tooling is **free**, target DB billed per **compute (vCore/DTU)**
Data-Only Transfer	**BACPAC Export/Import** (SSMS, Azure Portal, sqlpackage)	Export schema+data to .bacpac → Upload to Azure Blob → Import into Azure SQL	Smaller databases, <200GB typically	Azure Storage (per GB) + Azure SQL target billing
Large Data Files	**BULK INSERT / BCP / Azure Data Factory (ADF)**	Upload CSV/Parquet/Flat files → Use BULK INSERT/BCP locally, or ADF pipelines to Azure SQL	High-volume ETL, flat file ingestion, data warehouse loads	BCP/BULK free; **ADF billed per activity run + data movement (per DIU-hour)**
Ongoing Sync / Hybrid	**Transactional Replication / Change Data Capture (CDC) → Azure SQL MI**	Keep on-premises + cloud DB in sync during migration	Near-zero downtime cutover	Replication built-in (no extra cost), but compute billed in both source + target
Files → Blob → Synapse/SQL	**Azure Data Factory** or **Azure Synapse Pipelines**	Stage files in Blob/Data Lake → Map/Transform → Load into SQL pool	Big Data → Analytical SQL workloads	**ADF/Synapse pipeline billed per activity run & vCore usage**

Scenario	Tool / Service	Workflow Summary	Best For	Cost / Billing Unit
VM Lift-and-Shift	**Azure Migrate (VM)**	Move SQL Server VM → Run SQL as-is in Azure VM	Full server, unchanged	Billed per **Azure VM (per second)** + storage + SQL licensing
Backup/Restore Migration	**Native BACKUP/RESTORE to Azure Blob**	Backup DB to Blob Storage → Restore on Azure SQL Managed Instance	Large DBs where downtime is acceptable	Blob storage (per GB) + SQL MI billed per vCore

---

Key Exam B/R Pointers (DP-300 style)

- **DMS** is the official, fully-managed migration tool highlighted in exam content.
- **BACPAC** = logical export (schema + data) → good for smaller databases.
- **BACKUP/RESTORE to Blob** = physical migration, only supported with **SQL Managed Instance** (not Azure SQL DB).
- **BCP/BULK INSERT** = row-based, good for large flat file ingestion, but not schema migration.
- **Azure Data Factory** = cost appears when orchestrating ETL or moving large files.
- **Licensing & cost** always come down to the **target service billing model** (DTU vs vCore for Azure SQL, VM pricing for IaaS, DIU for ADF).

Indexes, Fragmentation, Rebuild, Reorganize

Great question — let’s dig in 👇

An **index seek** can occur on a non–primary key SQL column of a string (CHAR, VARCHAR, NVARCHAR) if a **nonclustered index** exists on that column, and the query optimizer determines it can efficiently use it.

Here’s what drives that decision:

🔑 Conditions for Index Seek on String Column

1. **Indexed Column**
   - The string column must have a **nonclustered index** (or be part of a composite index).
   - A seek can happen if the predicate references a leading column of the index.
2. **Sargable Predicate**
   - The query must be **search-argumentable (SARGable)**, meaning the condition can be transformed into a form that allows index navigation.
   - Examples that allow a seek:
   - WHERE LastName = 'Smith'
   - WHERE LastName >= 'M' AND LastName < 'N'
   - Non-SARGable (forces scan instead of seek):
   - WHERE LEFT(LastName, 1) = 'S'
   - WHERE UPPER(LastName) = 'SMITH'
3. **Collation and Comparisons**
   - Collation affects comparisons (case sensitivity, accent sensitivity).
   - If a function or conversion on the column breaks SARGability (e.g., WHERE CAST(Col AS NVARCHAR(50)) = ...), the optimizer can’t do a seek.
4. **Selectivity of Values**
   - The optimizer uses column **statistics** to decide between seek vs scan.
   - If the column has highly selective values (e.g., EmailAddress), a seek is very likely.
   - If most rows have the same value, it may still choose a scan.
5. **Matching Index Order / Prefix**
   - In a composite index (e.g., (LastName, FirstName)), you can seek on LastName alone, or LastName + FirstName, but not on FirstName alone.

✅ **Example**

CREATE NONCLUSTERED INDEX IX_Customers_LastName

ON Customers (LastName);

-- Seek happens:

SELECT * FROM Customers WHERE LastName = 'Doe';

-- Scan happens (non-SARGable):

SELECT * FROM Customers WHERE LEFT(LastName, 3) = 'Doe';

⚡ So the short version:
An **index seek** on a non–primary key string column happens when there is a nonclustered index on that column, and the query predicate is **SARGable** (direct comparison, range, LIKE with a fixed prefix). The optimizer then uses stats to decide if seeking is cheaper than scanning.

Decision Tree when SQL Server will use an **Index Seek** vs fall back to a **Scan** on a string column:

```
Start
 |
 ├── Does a nonclustered (or clustered) index exist
 |   on the string column?
 |      |
 |      ├─ No → Index Scan (or Table Scan)
 |      └─ Yes
 |
 ├── Is the WHERE predicate SARGable?
 |     (equality, range, LIKE 'abc%')
 |      |
```

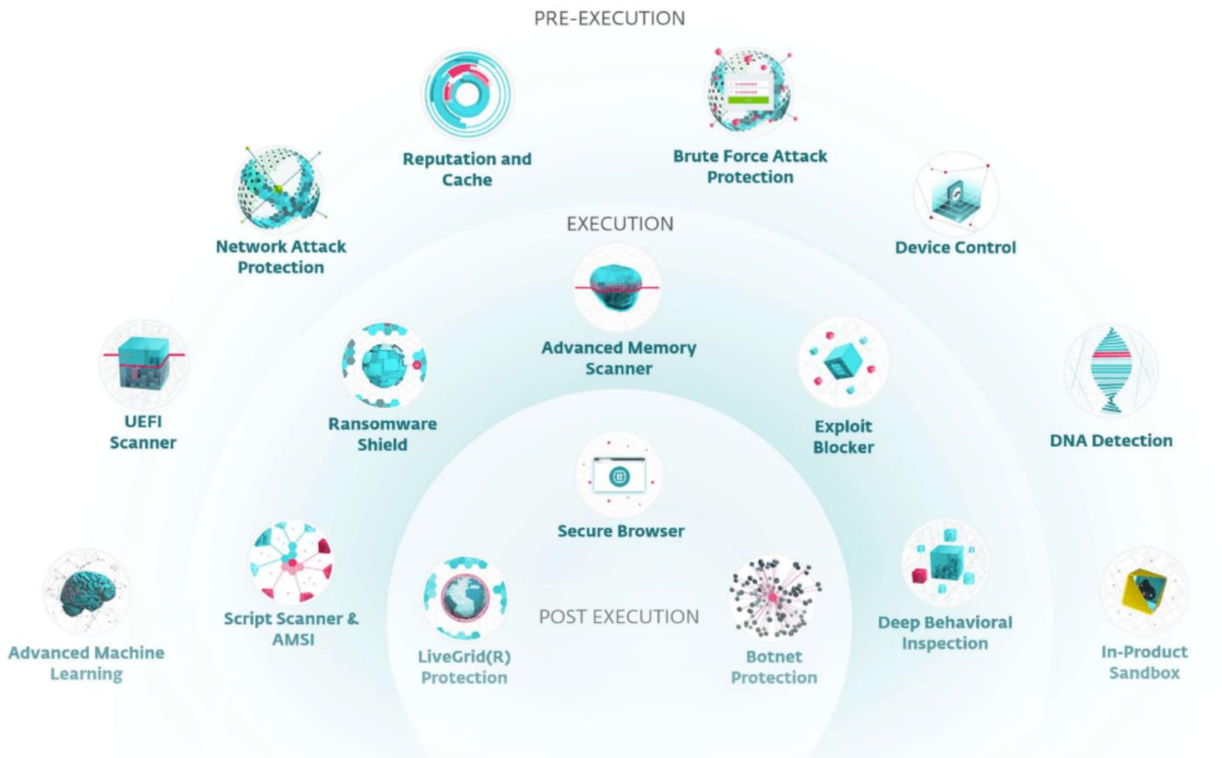
```
|       ├─ No (functions, wildcards at front, casts)
|       |       → Scan
|       └─ Yes
|
├── Is the index leading column used?
|       (single col or leftmost col in composite index)
|       |
|       ├─ No → Scan
|       └─ Yes
|
├── Do statistics show selective values?
|       |
|       ├─ Low selectivity (many rows match)
|       |       → Optimizer may choose Scan
|       └─ High selectivity (few rows match)
|               → Index Seek
|
└── Final Plan Choice → Seek or Scan
```

💡 Example outcomes:

- WHERE LastName = 'Smith' → Seek (if indexed, SARGable, selective).
- WHERE UPPER(LastName) = 'SMITH' → Scan (non-SARGable).
- WHERE FirstName = 'John' with index (LastName, FirstName) → Scan (not leading column).
- WHERE LastName LIKE 'Sm%' → Seek (prefix SARGable).
- WHERE LastName LIKE '%th' → Scan (wildcard at start).

Loose Notes & Diagrams

Items on the web



Working queries to try:

```
SELECT @@VERSION
SELECT * FROM sys.databases
SELECT * FROM sys.objects
SELECT * FROM sys.dm_os_schedulers
SELECT * FROM sys.dm_os_sys_info
SELECT * FROM sys.dm_os_process_memory –Not supported in Azure SQL Database
SELECT * FROM sys.dm_exec_requests
SELECT SERVERPROPERTY('EngineEdition')
SELECT * FROM sys.dm_user_db_resource_governance — Available only in Azure SQL Database and SQL Managed Instance
SELECT * FROM sys.dm_instance_resource_governance — Available only in Azure SQL Managed Instance
SELECT * FROM sys.dm_os_job_object — Available only in Azure SQL Database and SQL Managed Instance

GO
```

Hadoop data source

To query the data in your Hadoop data source, you must define an external table to use in Transact-SQL queries. The following steps describe how to configure the external table.

Step 1: Create a master key on database.

1. Create a master key on the database. The master key is required to encrypt the credential secret.

(Create a database scoped credential for Azure blob storage.)

Step 2: Create an external data source for Azure Blob storage.

2. Create an external data source with CREATE EXTERNAL DATA SOURCE.

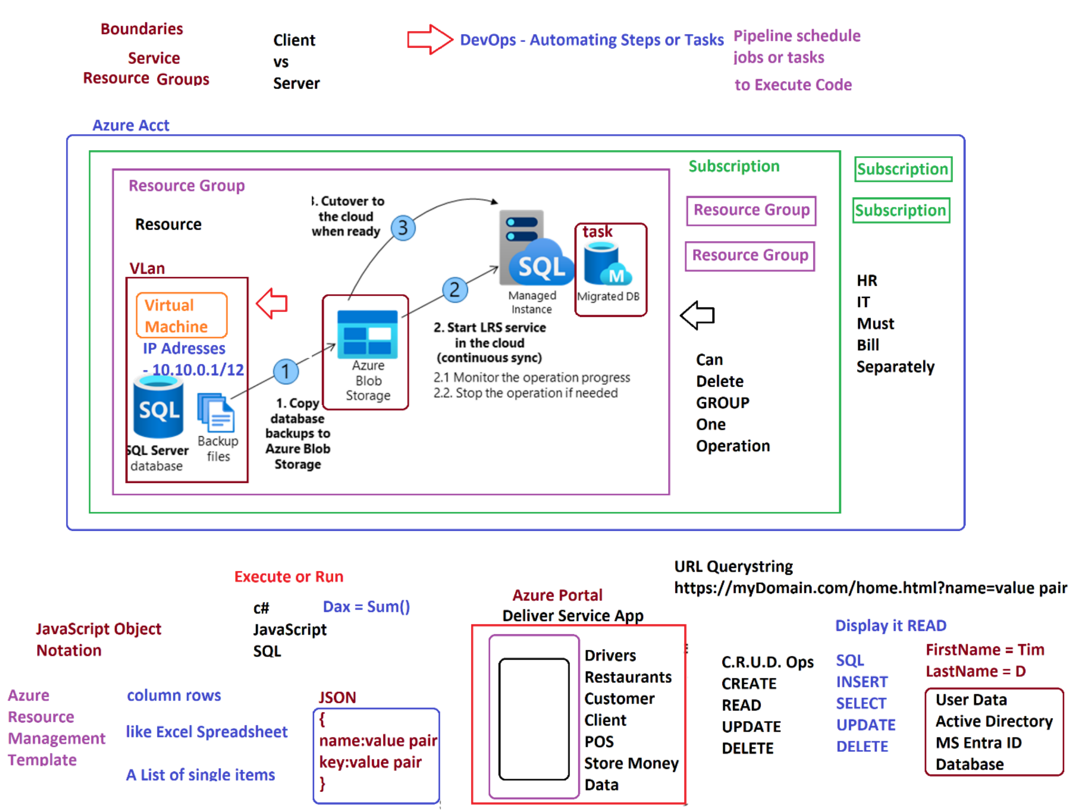
Step 3: Create an external file format to map the parquet files.

3. Create an external file format with CREATE EXTERNAL FILE FORMAT.

Step 4. Create an external table FactSalesOrderDetails

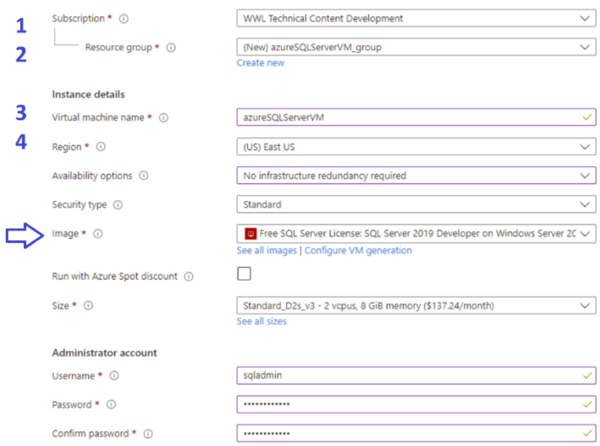
4. Create an external table pointing to data stored in Azure storage with CREATE EXTERNAL TABLE.

Reference: [Access external data: Azure Blob Storage - PolyBase - SQL Server | Microsoft Learn]()



portal fill out form JSON

INSERT - NEW RECORDS
UPDATE - EXISTS

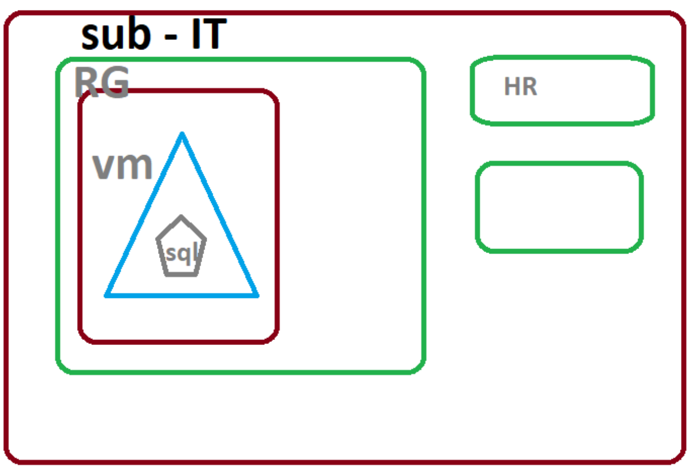


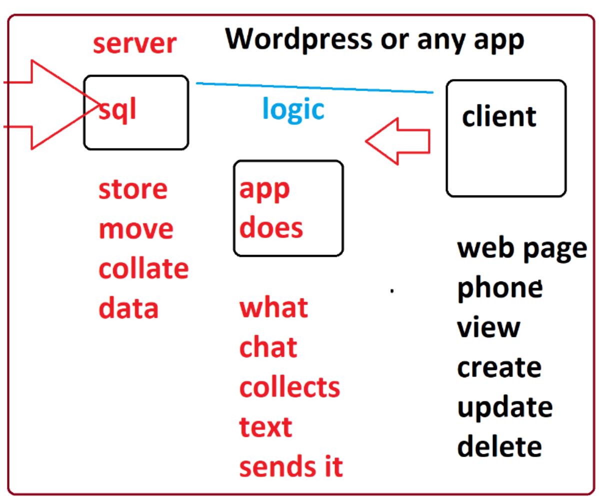
5. On the **Create a virtual machine** page, enter the following information:

- **Subscription:** <Your subscription>
- **Resource group:** <Your resource group>
- **Virtual machine name:** azureSQLServerVM
- **Region:** <your local region, same as the selected region for your resource group>
- **Availability Options:** No infrastructure redundancy required
- **Image:** Free SQL Server License: SQL 2019 Developer on Windows Server 2022 - Gen1
- **Azure spot instance:** No (unchecked)
- **Size:** Standard *D2s_v3* (2 vCPUs, 8 GiB memory). You may need to select the "See all sizes" link to see this option)
- **Administrator account username:** sqladmin
- **Administrator account password:** pwd!DP300lab01 (or your own password that meets the criteria)
- **Select inbound ports:** RDP (3389)
- **Would you like to use an existing Windows Server license?:** No (unchecked)

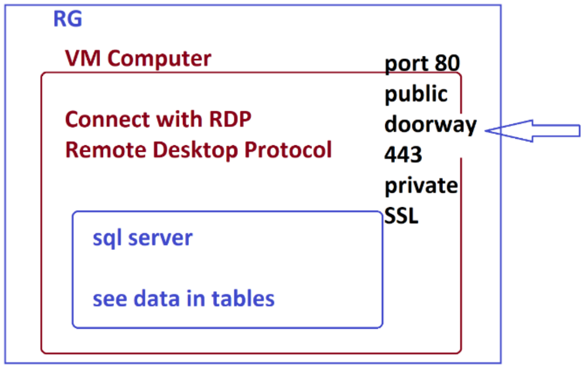
Make note of the username and password for later use.

separate billing





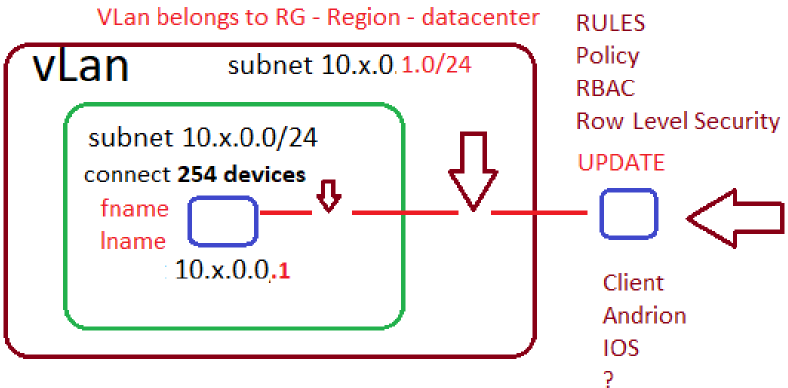
Electricity - + Positive - Negative electron	Type Code Compiles Interpret
Boolean True or False 1 0	Turns Text to Binary 1 - 0

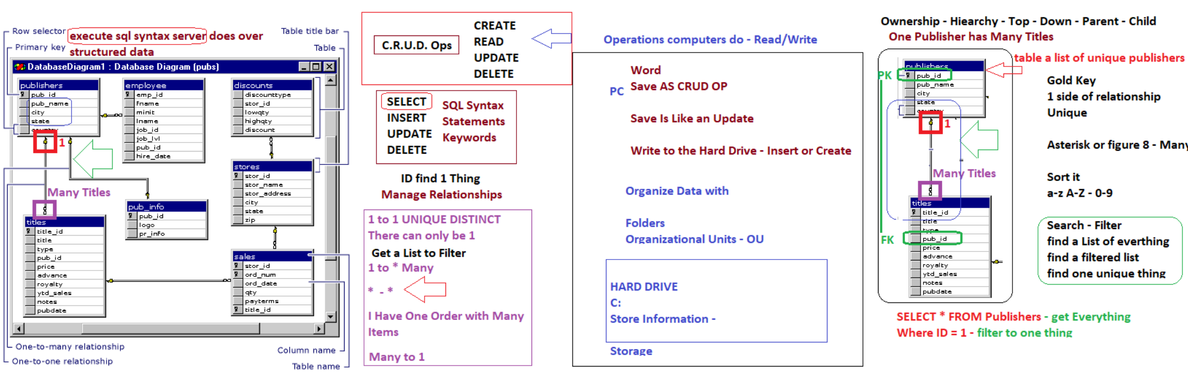


RG - Region - Zone - Set

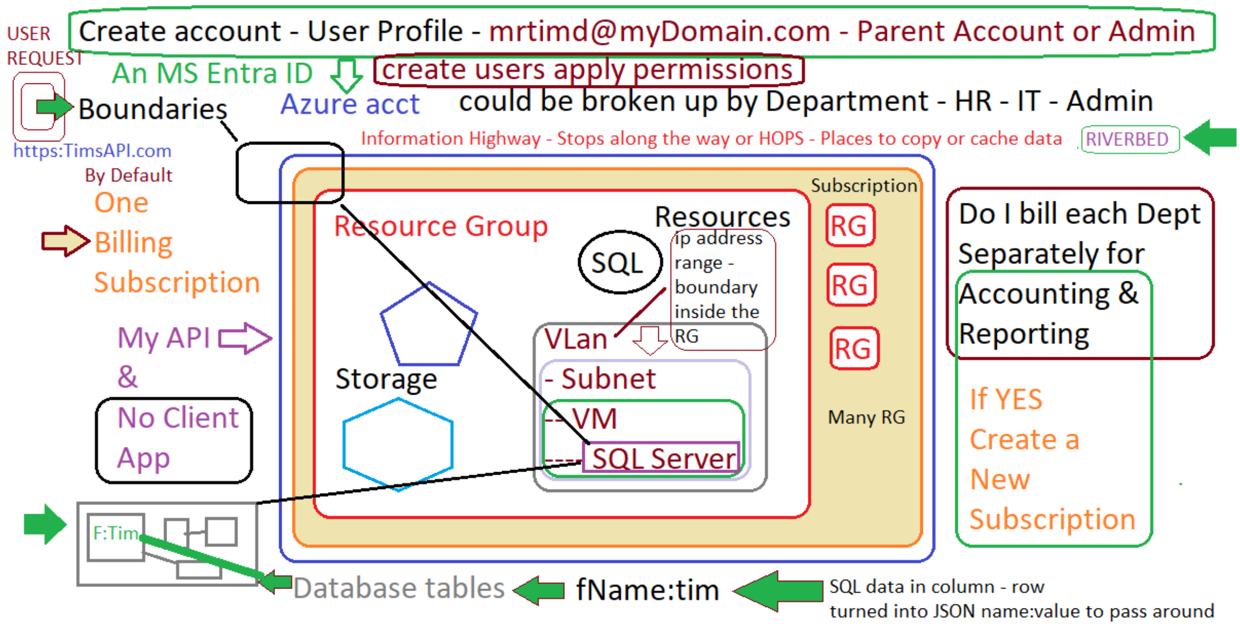
Server Side - SQL Database

VLan belongs to RG - Region - datacenter





	A	B	C
5	Publishers Table		
6	Primary Key ID	Name	Foreign Key ID
7	1	Amadis (LOC) Karen	1
8	2	Armas (LOC) Dayana	1
9	3	Arnao (LOC) Nelly	2


USER REQUEST
Create account - User Profile - mrtimd@myDomain.com - Parent Account or Admin
An MS Entra ID
create users apply permissions
Boundaries
Azure acct
could be broken up by Department - HR - IT - Admin
https:TimsAPI.com
Information Highway - Stops along the way or HOPS - Places to copy or cache data
RIVERBED
By Default
One
Billing
Subscription
Subscription
Resource Group
Resources
SQL
ip address range - boundary inside the RG
RG
RG
RG
Do I bill each Dept Separately for Accounting & Reporting
My API
&
No Client App
Storage
VLan
- Subnet
-- VM
--- SQL Server
Many RG
If YES Create a New Subscription
F:Tim
Database tables
fName:tim
SQL data in column - row
turned into JSON name:value to pass around

📘 DP-300 Full Practice Quiz (30 Questions)

🔷 Section 1: Easy (1–10)

Q1.
Which Azure service provides a fully managed SQL Server instance with near 100% compatibility?
A. Azure SQL Database
B. Azure SQL Managed Instance
C. Azure Synapse Analytics
D. SQL Server on VM

Q2.
Which purchasing model uses DTUs?
A. Hyperscale
B. vCore-based
C. Elastic Pools
D. DTU-based

Q3.
Which service automatically applies security patches in Azure SQL Database?
A. Microsoft Defender for SQL
B. Azure Policy
C. Platform-as-a-Service (PaaS) layer
D. Transparent Data Encryption (TDE)

Q4.
Which SQL Server feature is **NOT supported** in Azure SQL Database?
A. Partitioned Tables
B. Cross-database transactions
C. Columnstore indexes
D. In-Memory OLTP

Q5.
Which command retrieves the current Service Tier and Performance Level in Azure SQL Database?
A. SELECT * FROM sys.dm_exec_requests
B. SELECT * FROM sys.database_service_objectives
C. DBCC CHECKDB
D. SELECT * FROM sys.dm_exec_sessions

Q6.
Azure SQL Database is backed up automatically. How long are backups retained by default in the Standard tier?
A. 7 days
B. 14 days
C. 30 days
D. 35 days

Q7.
Which feature encrypts data at rest in Azure SQL Database?
A. Always Encrypted
B. Transparent Data Encryption (TDE)
C. Dynamic Data Masking
D. TLS/SSL

Q8.
Which tool provides **assessment reports** before migrating SQL Server to Azure?
A. Data Migration Assistant (DMA)
B. SQL Server Profiler
C. Extended Events
D. Query Store

Q9.
What must you configure to allow **Azure AD authentication** in Azure SQL Database?
A. Firewall rules only
B. Azure AD Admin at the server level
C. Database role membership
D. SQL logins only

Q10.
Which service tier is required for **Hyperscale databases**?
A. DTU-based Basic
B. vCore-based General Purpose or Business Critical
C. Elastic Pool Standard
D. Premium DTU

🔷 Section 2: Medium (11–20)

Q11.
Which type of **replication** is supported in Azure SQL Database?
A. Transactional Replication (Subscriber only)
B. Peer-to-Peer Replication
C. Merge Replication
D. Snapshot Replication

Q12.
Which DMV helps you identify the **top resource-consuming queries** in Azure SQL Database?
A. sys.dm_db_index_usage_stats
B. sys.dm_exec_query_stats
C. sys.dm_exec_requests
D. sys.dm_db_resource_stats

Q13.
You need to configure **transparent failover** between primary and secondary databases in different regions. Which option should you use?
A. Geo-Restore
B. Active Geo-Replication
C. Failover Groups
D. Always On Availability Groups

Q14.
Which service tier guarantees **low-latency high I/O performance** and is optimized for business-critical apps?
A. General Purpose
B. Hyperscale
C. Business Critical
D. Elastic Pool

Q15.
Which command checks the last backup date for a database in Azure SQL Managed Instance?
A. RESTORE VERIFYONLY
B. SELECT * FROM msdb..backupset
C. SELECT * FROM sys.dm_database_backups
D. DBCC CHECKDB

Q16.
Which Azure monitoring service provides **built-in alerts and dashboards** for SQL performance?
A. Azure Monitor
B. Log Analytics
C. SQL Profiler
D. Performance Monitor

Q17.
Which feature lets Azure SQL Database **automatically force last known good plans** for queries?
A. Query Store
B. Automatic Plan Correction
C. Extended Events
D. Hints

Q18.
Which method provides **point-in-time restore** for Azure SQL Database?
A. Geo-restore
B. Long-term backup retention (LTR)
C. Automatic backups
D. Always On

Q19.
Which tool is best suited to run **Kusto Query Language (KQL)** against diagnostic logs from Azure SQL Database?
A. SQL Server Management Studio
B. Azure Data Explorer / Log Analytics
C. Query Store
D. SQL Profiler

Q20.
Which security feature replaces sensitive values with placeholders at query time?
A. Always Encrypted
B. Transparent Data Encryption
C. Dynamic Data Masking
D. Row-Level Security

🔷 Section 3: Hard (21–30)

Q21.
You need to migrate a **10TB on-premises database** with minimal downtime to Azure SQL Managed Instance. Which approach is recommended?
A. BACPAC export/import
B. Transactional Replication
C. Log Shipping + Database Migration Service (DMS)
D. Copy Database Wizard

Q22.
Which isolation level provides the **least concurrency issues** in Azure SQL Database?
A. Read Committed
B. Read Uncommitted
C. Serializable
D. Snapshot

Q23.
Which DMV helps detect **blocking sessions** in Azure SQL Database?
A. sys.dm_exec_sessions
B. sys.dm_exec_requests
C. sys.dm_tran_locks
D. sys.dm_exec_connections

Q24.
You are designing a **multi-tenant SaaS application**. Which Azure SQL Database option provides **best cost efficiency**?
A. Single Database – vCore provisioned
B. Single Database – Hyperscale
C. Elastic Pool
D. SQL Managed Instance

Q25.
Which tool allows **notebook-style experiences** with T-SQL, Python, and PowerShell for DBAs?
A. SQL Server Management Studio (SSMS)
B. Azure Data Studio
C. SQL Profiler
D. Log Analytics

Q26.
Which method is best for encrypting **data in transit** for Azure SQL Database connections?
A. TDE
B. Always Encrypted
C. TLS/SSL
D. Row-Level Security

Q27.
In Azure SQL Database, what's the **max storage** supported in the Hyperscale tier?
A. 4 TB
B. 8 TB
C. 100 TB
D. Unlimited

Q28.
Which DMV shows **wait statistics** in Azure SQL Database?
A. sys.dm_exec_query_stats
B. sys.dm_os_wait_stats
C. sys.dm_tran_session_transactions
D. sys.dm_exec_requests

Q29.
You configure **automatic tuning** in Azure SQL Database. Which settings can it adjust automatically?
A. Index creation and dropping, force last good plan
B. TempDB file growth, statistics updates
C. Partitioning strategies
D. Always Encrypted policies

Q30.
Which option allows retaining backups for **up to 10 years** in Azure SQL Database?
A. Geo-Replication
B. Long-Term Retention (LTR)
C. Active Failover Groups
D. Point-in-Time Restore

☑ Answer Key

Easy (1–10):

1-B, 2-D, 3-C, 4-B, 5-B, 6-D, 7-B, 8-A, 9-B, 10-B

Medium (11–20):

11-A, 12-B, 13-C, 14-C, 15-C, 16-A, 17-B, 18-C, 19-B, 20-C

Hard (21–30):

21-C, 22-D, 23-B, 24-C, 25-B, 26-C, 27-C, 28-B, 29-A, 30-B

End Practice Q's & Key

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End dp-300